A larger budget, a board that forgets faster or slower

Simulation 2 (asynchronous activation), setup `task003-symmetric`: three variants against the gold standard

LLM-free simulator; report of 19 October 2026; runs of 15, 16 and 18 October 2026

Same sections, figures and equations as the gold-standard report of 16 October, with the four cases side by side.

How the simulator works. The world, `task_003`, has three candidate allocations; A0 is the correct one. Each of 24 agents is given one true fact and reasons as an exact Bayesian: its *belief* is the exact probability of A0, A1 and A2 given the facts it currently remembers. There are no rounds. Each agent acts at random times, on average once per time unit: it reads q posts drawn at random from the W most recent posts ($W = 48$ in the gold standard) on a shared *board* (never its own), adds their facts to its memory, recomputes its belief, *votes* by probability matching (a vote drawn from its belief) and posts one remembered fact that supports its vote. Every remembered fact is forgotten after a random lifetime (on average a fraction ρ survives one time unit; reading it again renews it). A *controller* with a target, A0 (truth) or A2 (false), acts at random times at rate λ_c until $t = 30$: it reads q_c recent agent posts and, unless the *vote gate* (at least 75% of them from agents voting for its target) or the *proof stop* (its target already proved) silences it, posts the one fact from its pool that best supports its target, at most B messages per episode ($B = 30$ in the gold standard). Agents continue alone until $t = 40$. Every episode is run three times with the same random draws: without a controller (*silent*), with the controller targeting A0 and with the one targeting A2.

The facts at the start. All facts are true. The agents hold, together, a proof of A0; no single agent does, and neither pool can prove its target (Table 1).

Table 1: Facts available at $t = 0$ and what each set implies, $P(\text{A0}, \text{A1}, \text{A2} \mid \text{facts})$.

set of facts	facts	$P(\text{A0}, \text{A1}, \text{A2})$
agents' facts, all together (24 agents, one fact each)	18 distinct	(1, 0, 0): proves A0
A0 pool (controller with target A0)	12, of which 10 new to the agents	(0.963, 0.037, 0)
A2 pool (controller with target A2)	12, of which 10 new to the agents	(0.114, 0, 0.886)
agents' facts + either pool	28 distinct	(1, 0, 0)

The three variants. Each changes one rule of the gold standard and keeps everything else (Table 2).

- **Budget.** In the gold standard the controller may post at most $B = 30$ messages before $t = 30$; from $\lambda_c = 2$ upward it uses almost all of them, and at $\lambda_c = 8$ it runs out near $t = 4$. With $B = 100$ it can keep posting until $t = 30$ for $\lambda_c \lesssim 3$ and runs out near $t = 12.5$ at $\lambda_c = 8$; the vote gate can still silence it. This separates the rate from the total effort.
- **Board window.** Nothing is ever deleted from the board, but every reader (agent or controller) draws only from the W newest posts it may read. The 24 agents post about 24 times per time unit, so in the gold standard ($W = 48$) a post stays readable for about 2 time units unless someone posts its fact again. With $W = 24$ it stays about 1 time unit (a board that forgets fast); with $W = 240$ about 10 (a board that forgets slowly). The window also changes what the controller reads, so with $W = 240$ its vote gate sees older votes.

Table 2: The four cases compared in this report. All: $q_c = 12$, $\rho \in \{1, 0.75, 0.5, 0.25\}$, $q \in \{3, 6, 12\}$, all nine rates, gate on and off, 1,000 episodes per setting.

case	budget B	window W	a post stays readable	run
gold standard	30	48	≈ 2 time units	15–16 October (`*_fine`)
$B = 100$	100	48	≈ 2	18 October (`*_budget100`)
$W = 24$	30	24	≈ 1	18 October (`*_window24`)
$W = 240$	30	240	≈ 10	18 October (`*_window240`)

The runs. The gold standard's settings at $q_c = 12$ and $\rho \in \{1, 0.75, 0.5, 0.25\}$ (from the 15–16 October runs) and the three variants on the same grid: posts read by an agent $q \in \{3, 6, 12\}$; rate $\lambda_c \in \{0.5, 1, 2, 3, 4, 5, 6, 7, 8\}$; vote gate on and off; proof stop on; 1,000 episodes per setting (444 settings per case). Unless stated otherwise:

$q = 6$, $q_c = 12$, $\lambda_c = 1$, gate on. Where the budget cannot matter, the $B = 100$ run reproduces the gold standard exactly: all silent settings, and every setting in which the gold-standard controller never reached 30 messages (113 settings).

How this report is organised. Part A (Section 2) compares the gold standard with $B = 100$; Part B (Section 3) compares it with $W = 24$ and $W = 240$. Each part has the gold standard's sections. Only the false controller (target A2) is shown: the controller targeting A0 changes by at most 0.03 in every variant. Readability I and efficiency η still use both targets, as defined in Section 1.

Summary. The gold standard's main results hold in all three variants: full memory makes the swarm immune, forgetting makes it steerable, and reading more protects when agents remember and helps the controller when they forget. **A budget of 100 messages lets fast controllers keep gaining** (Part A): up to +0.13 at $\rho \le 0.5$, with each extra message worth about half as much; the 30-message cap is what made the rate look irrelevant to the end point. **The board's memory protects like the agents' memory, but less** (Part B): posts that stay readable for 10 time units instead of 2 lower the false controller's gain by 0.06–0.14 at $\rho \le 0.5$; posts readable for 1 time unit raise it by up to 0.09.

1 Quantities

Votes. With $n_{a,t}$ the number of agents whose last vote is answer a at time t and N_t the number of agents who have voted,

$$x_{a,t} = n_{a,t}/N_t. \tag{1}$$

At $t = 0$ nobody has voted and we set $x_{a,0} = 0$; all agents have voted by $t \approx 5$. (The 9–14 October reports divided by 24, which counted agents who had not yet acted as voting against the target; this inflated the apparent rise of the shares and the apparent bias of the controller's sensing at early times.)

Paired susceptibility. Following `BLACKBOARD_OBSERVABLES_AND_EFFICIENCIES` (§3), for a controller policy π and an answer a,

$$\chi^{(a)}_{\pi,t} \equiv G^{(a)}_t(\pi) = \mathbb{E}\big[x^{\pi}_{a,t} - x^{\text{base}}_{a,t}\big], \tag{2}$$

the mean over episodes of the difference between the controlled run and the silent run of the same episode. We report it on the controller's own target: $G_t \equiv G^{(0)}_t(\pi^+)$ for the controller targeting A0 and $G_t \equiv G^{(2)}_t(\pi^-)$ for the one targeting A2.

Target and state. The target $Z \in \{\text{A0}, \text{A2}\}$ is drawn by a fair coin. The state X at time t is the vote of a randomly chosen agent, $X \in \{\text{A0}, \text{A1}, \text{A2}\}$. Each episode e of arm s (silent 0, target A0 +, target A2 −) contributes its vote shares $y^{(e)}_s(x) = x_{x,t}$, and

$$P_s(x) = \frac{1}{E}\sum_{e=1}^{E} y^{(e)}_s(x), \qquad E = 1000. \tag{3}$$

Readability and efficiency. With $Q = \frac{1}{2}(P_+ + P_-)$ and $\mathrm{KL}(p\|q) = \sum_x p(x)\log_2[p(x)/q(x)]$,

$$I(Z;X) = \tfrac{1}{2}\,\mathrm{KL}(P_+\|Q) + \tfrac{1}{2}\,\mathrm{KL}(P_-\|Q), \tag{4}$$

the information the state carries about the target (at most 1 bit). The controllers' total change from silence splits exactly into a directed and an undirected part,

$$C = \tfrac{1}{2}\,\mathrm{KL}(P_+\|P_0) + \tfrac{1}{2}\,\mathrm{KL}(P_-\|P_0) = I(Z;X) + \mathrm{KL}(Q\|P_0), \tag{5}$$

and the efficiency is the directed share,

$$\eta = I(Z;X)/C. \tag{6}$$

Silence sometimes never shows a value of X that a controller reaches, which would make C infinite. Before computing any KL, every arm's distribution therefore receives the same pseudo-count of $\alpha = 0.5$ episodes per value of X (K values),

$$\tilde{P}_s(x) = \frac{E\,P_s(x) + \alpha}{E + \alpha K}. \tag{7}$$

With $\alpha = 1$ instead, η changes by at most 0.004; without it, by at most 0.005 where C is finite. η is not defined when $C < 0.01$ bits (no measurable change); such values are left blank. Equation (5) holds in the data to 4×10^{-15} bits.

Cost. F_t is the number of facts the controller has posted up to t (mean per episode); the gain per fact is G_t/F_t, and the marginal susceptibility is the slope between neighbouring rates,

$$\frac{\Delta G}{\Delta \lambda_c} = \frac{G(\lambda_c^{(k+1)}) - G(\lambda_c^{(k)})}{\lambda_c^{(k+1)} - \lambda_c^{(k)}}. \tag{8}$$

Episode e uses the same random draws at every rate, so the slope is estimated from per-episode differences, $[x^{(e)}_{\text{target},t}(\lambda_c^{(k+1)}) - x^{(e)}_{\text{target},t}(\lambda_c^{(k)})]/(\lambda_c^{(k+1)} - \lambda_c^{(k)})$, whose mean is Eq. (8) and whose standard error gives the 95% intervals (typically ±0.01–0.03, about 30% narrower than for two independent means).

2 Part A: a larger budget ($B = 100$)

The gold standard ($B = 30$) against $B = 100$; the window is $W = 48$ in both.

2.1 Votes over time

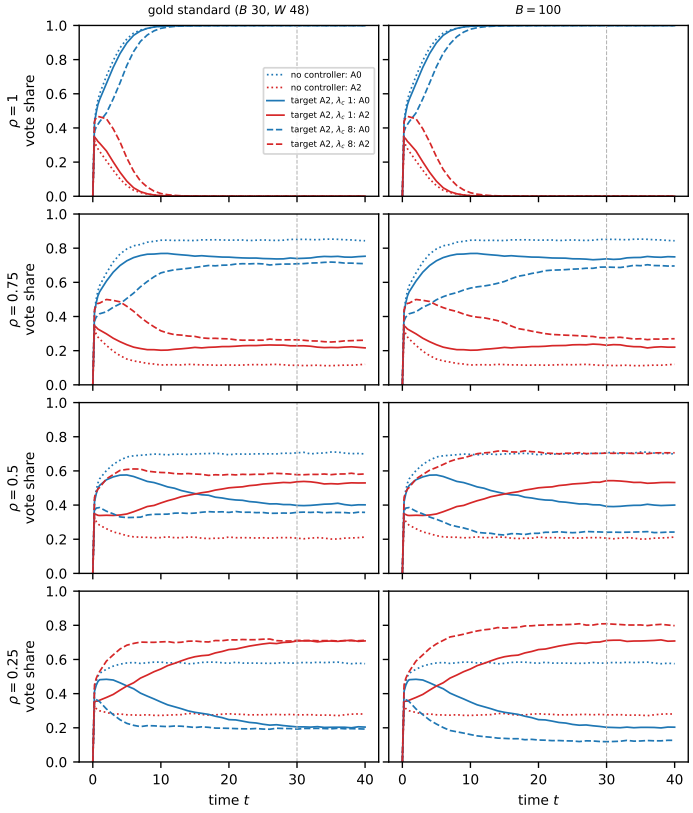


Figure 1: Share of agents voting A0 (blue) and A2 (red), Eq. (1): without a controller (dotted) and with the false controller (target A2) at $\lambda_c = 1$ (solid) and $\lambda_c = 8$ (dashed). Rows: memory ρ. $q = 6$, $q_c = 12$, gate on. At $t = 0$ nobody has voted and every share is 0.

- **At $\lambda_c = 1$ nothing changes**: the controller rarely reaches 30 messages, so the curves equal the gold standard's.
- **At $\lambda_c = 8$ the false controller wins more votes where the swarm forgets**: A2 on $t = 30$ goes from 0.58 to 0.70 ($\rho = 0.5$) and from 0.71 to 0.81 ($\rho = 0.25$); at $\rho = 0.75$ from 0.26 to 0.28; at $\rho = 1$ it is 0 in both. The results persist to $t = 40$.

2.2 Susceptibility over time, and per message

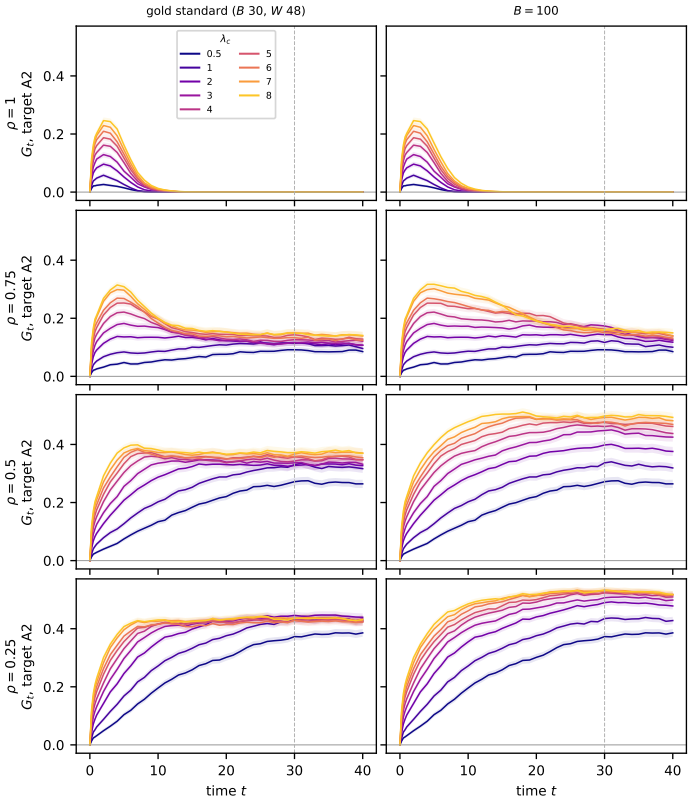


Figure 2: Paired susceptibility G_t, target A2, Eq. (2), one curve per rate λ_c (bands: 95% intervals). Rows: ρ.

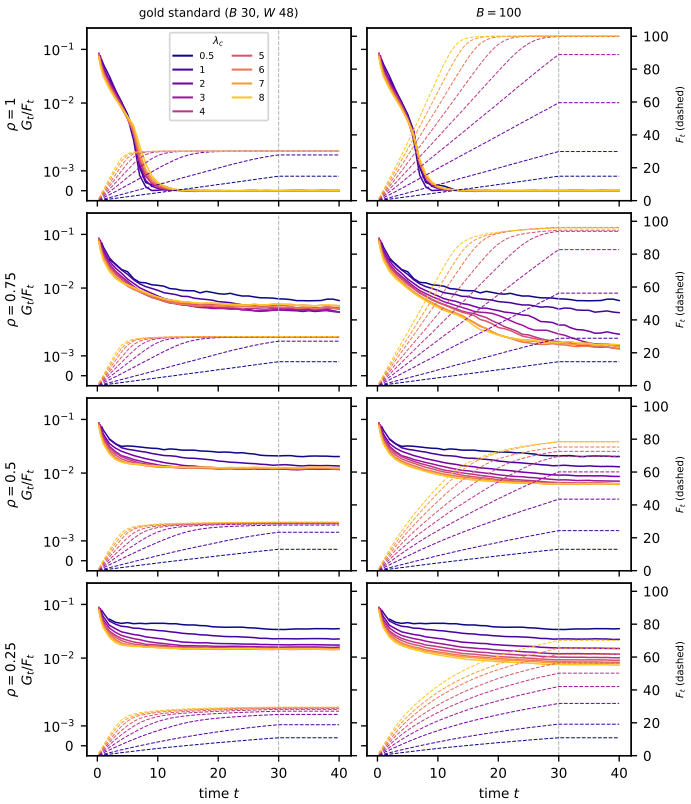


Figure 3: Gain per message G_t/F_t (solid, left axis, log scale) and messages posted F_t (dashed, right axis), target A2, one colour per rate.

- **Fast controllers keep gaining after the gold standard's budget would have run out** (Figure 2): on $t = 30$, G ranges over the rates from 0.27 to 0.50 at $\rho = 0.5$ (gold standard 0.27–0.37) and from 0.37 to 0.53 at $\rho = 0.25$ (0.37–0.45); at $\rho = 0.75$ it rises by at most 0.06, at $\rho = 1$ not at all.
- **Readability follows** (Figure 5, middle): on $t = 30$, $\rho = 0.5$, $I = 0.15$–0.38 bits against 0.15–0.27. **The efficiency drops a little** (right): $\eta = 0.76$–0.91 against 0.82–0.93 at $\rho < 1$.
- **Each extra message is worth about half as much** (Figure 3): on $t = 30$, $\lambda_c = 8$, 0.0063 per message against 0.0126 at $\rho = 0.5$ (78 messages against 29), 0.0076 against 0.0148 at $\rho = 0.25$.

2.3 Phase diagrams

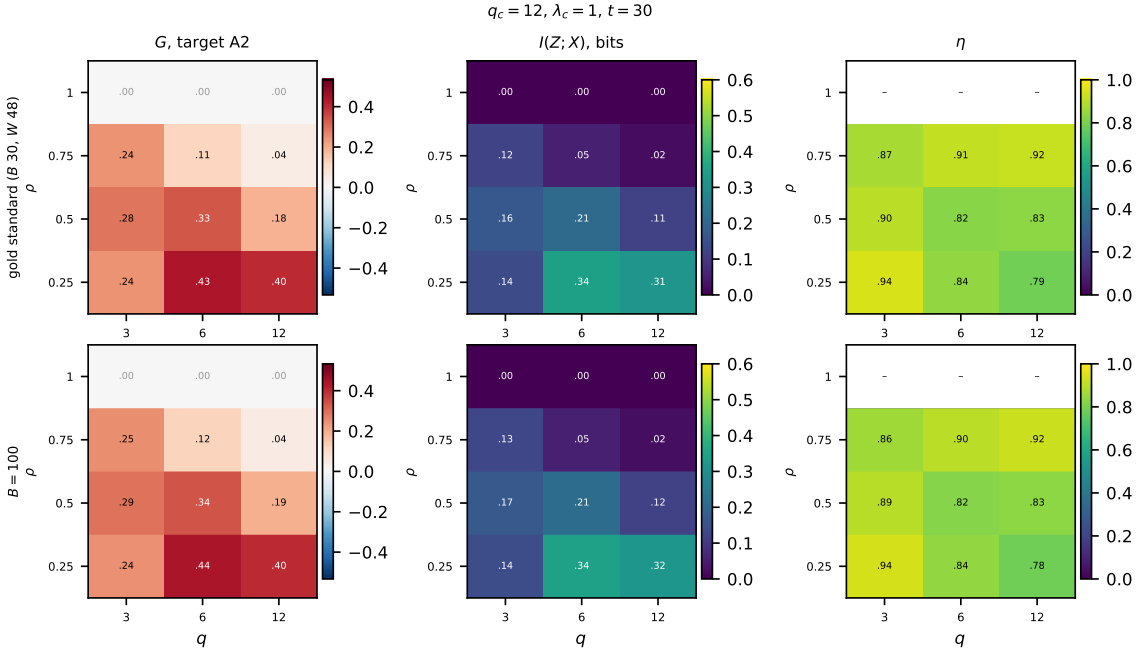


Figure 4: Susceptibility (target A2; grey number: 95% interval includes 0), readability and efficiency on $t = 30$ over memory ρ and the posts each agent reads q. $q_c = 12$, $\lambda_c = 1$, gate on.

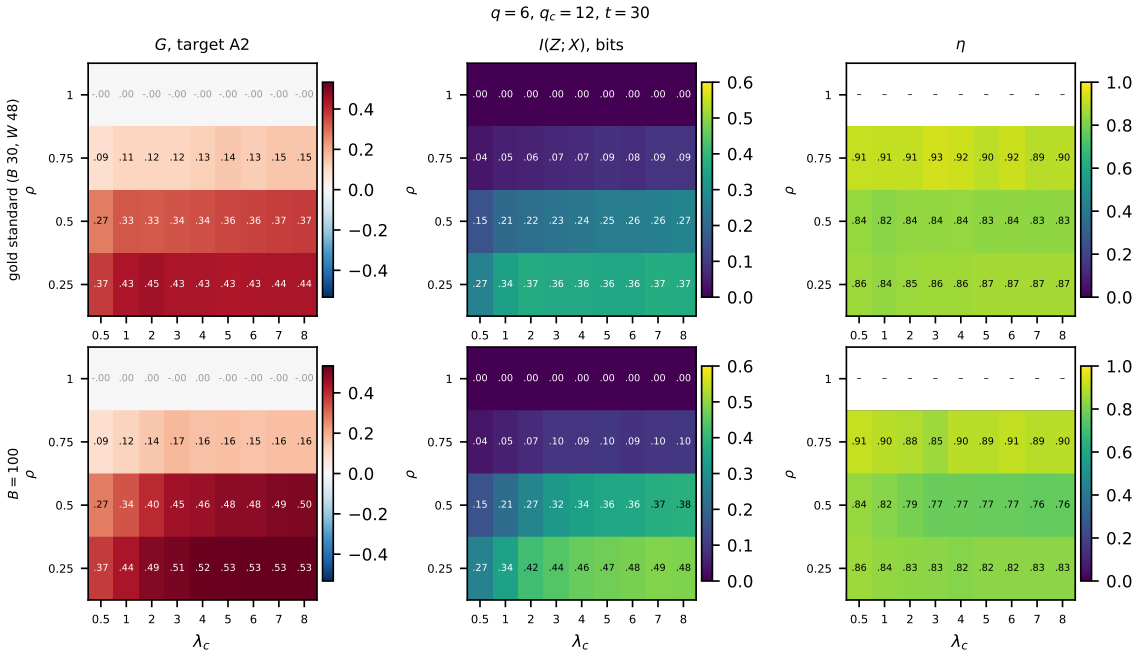


Figure 5: The same over memory ρ and the rate λ_c. $q = 6$, $q_c = 12$.

- **At $\lambda_c = 1$ the reading reversal is unchanged** (Figure 4): reading more protects at $\rho = 0.75$ (G 0.25 $\rightarrow$ 0.04 from $q = 3$ to 12) and helps the controller at $\rho = 0.25$ (0.24 $\rightarrow$ 0.40).
- **The budget adds a rate gradient** (Figure 5): from $\lambda_c = 1$ to 8, G grows by 0.16 ($\rho = 0.5$) and 0.10 ($\rho = 0.25$), against 0.04 and 0 in the gold standard.

2.4 Susceptibility against the rate, against messages, and its slope

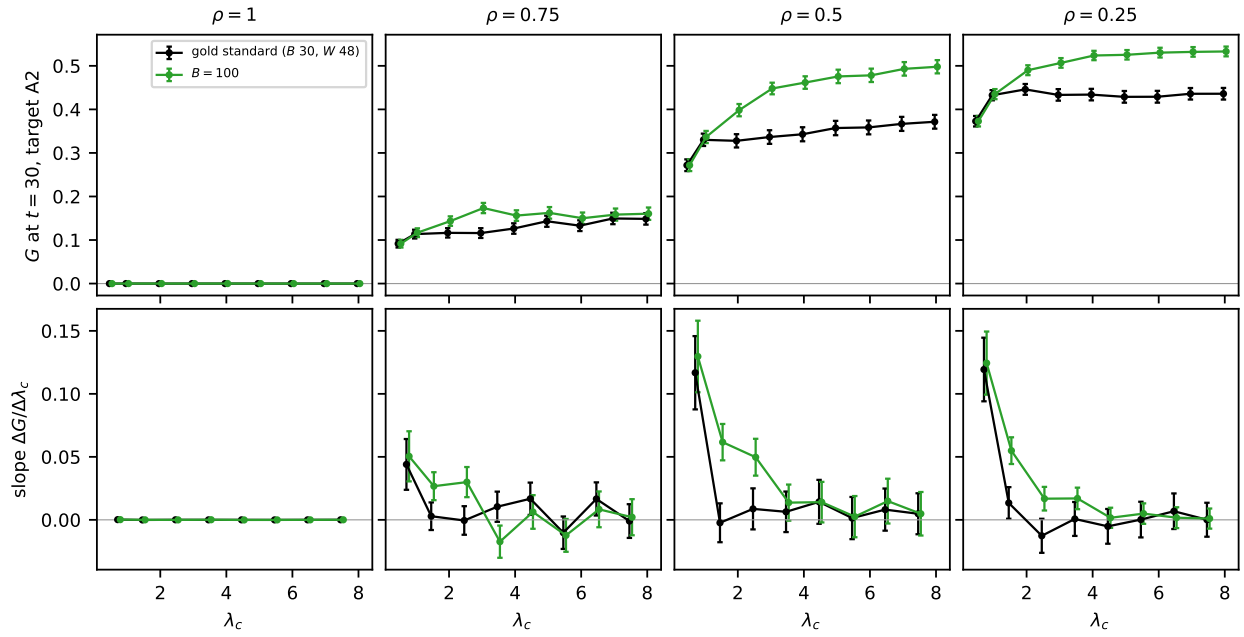


Figure 6: Top: G on $t = 30$, target A2, against the rate (bars: 95% intervals). Bottom: the slope between neighbouring rates, Eq. (8), with paired 95% intervals. $q = 6$, $q_c = 12$, gate on.

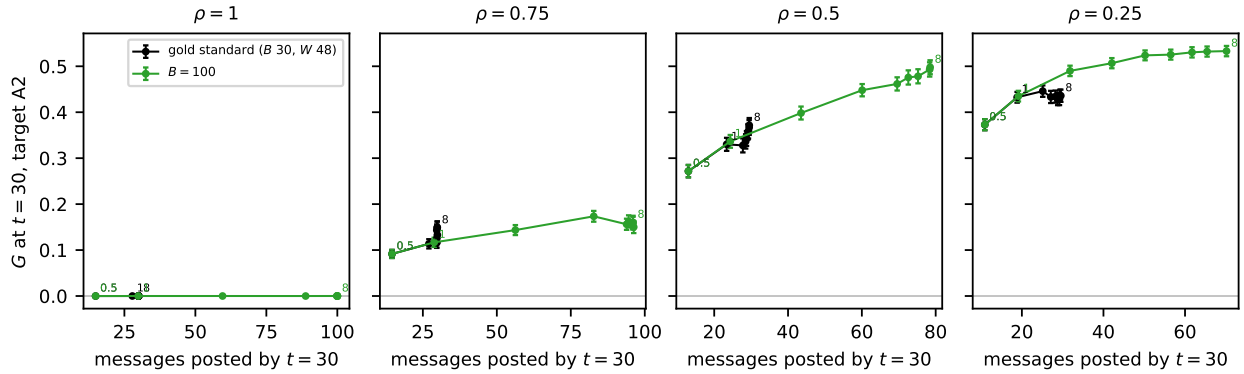


Figure 7: G on $t = 30$, target A2, against the messages posted by then; one point per rate, joined in rate order (labels: $\lambda_c = 0.5$, 1, 8); bars: 95% intervals. Read it like Figure 9 of the 19 October gold-standard copy: moving right is more messages, moving up is more gain.

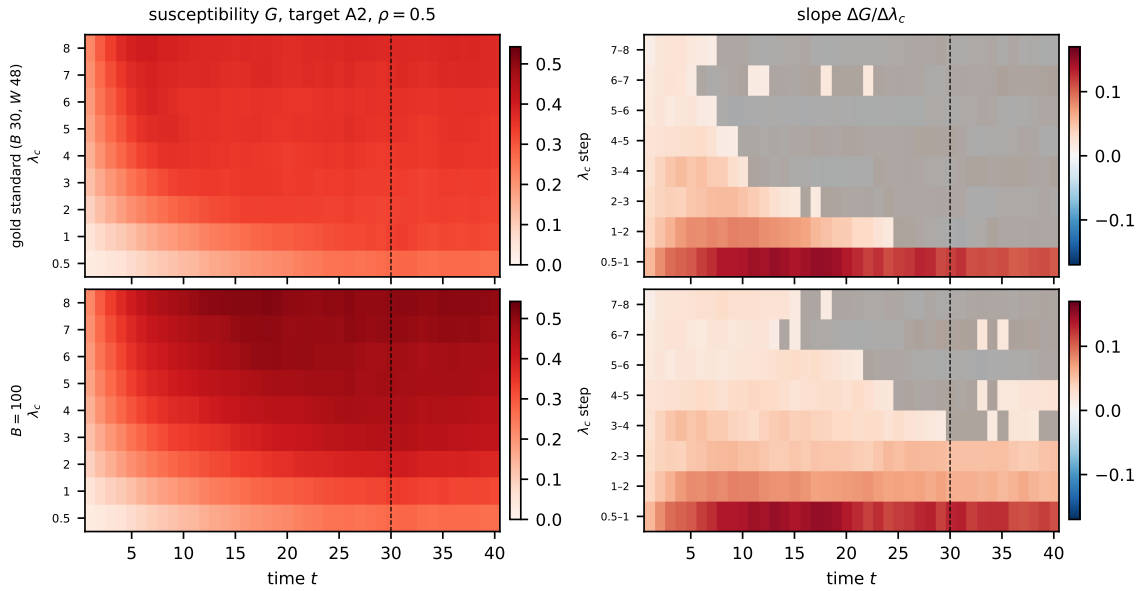


Figure 8: G, target A2 (left), and its slope between neighbouring rates (right) over time and rate, $\rho = 0.5$. Faded cells: the paired 95% interval of the slope includes 0.

- **With $B = 100$ more rate keeps paying on** $t = 30$ (Figure 6): at $\rho = 0.5$ the steps up to $\lambda_c = 3$ are significant (+0.13, +0.06, +0.05 per unit of rate); in the gold standard only the first (+0.12).
- **The budget moves the controller along one curve** (Figure 7): with 30 messages the gold standard's points pile up at the budget; with 100 they continue the same rising curve, from 0.33 (24 messages) to 0.50 (78 messages) at $\rho = 0.5$.
- **Over time** (Figure 8): in the gold standard the steps above $\lambda_c = 1$ fade between $t \approx 5$ and 25; with $B = 100$ those up to $\lambda_c = 3$ stay positive to $t = 30$.
- Gate off minus on: -0.01 to $+0.04$, as in the gold standard.

2.5 Who holds the proof

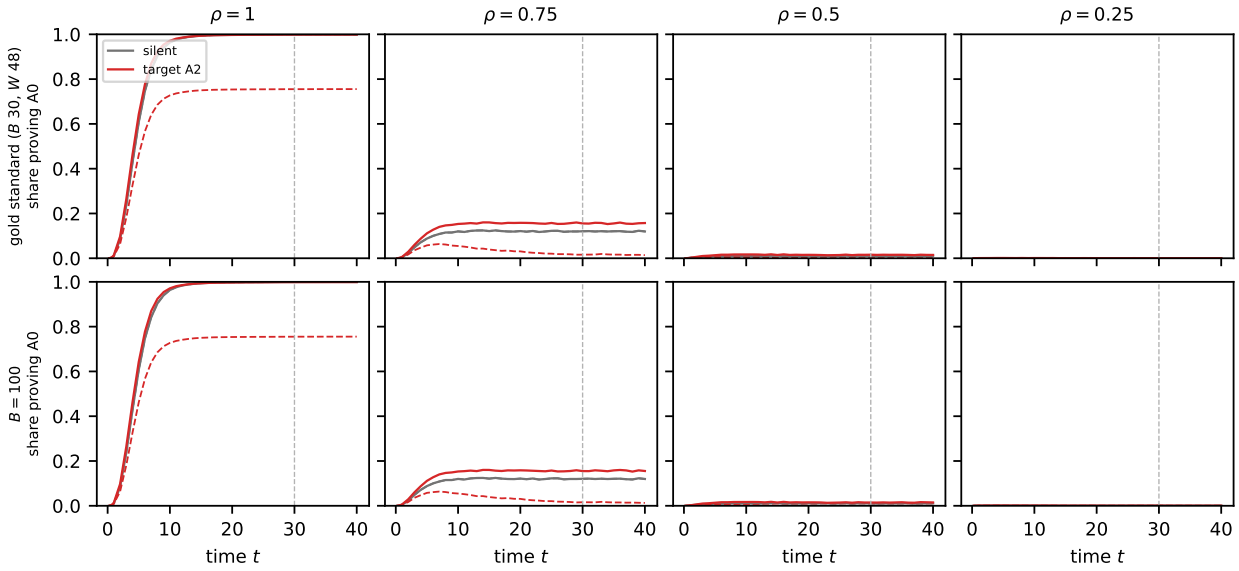


Figure 9: Share of agents whose remembered facts prove A0 (solid) and who prove it from their original facts only (dashed), without a controller (grey) and with the false controller at $\lambda_c = 1$ (red); columns ρ.

At $\lambda_c = 1$ the budget does not change who holds the proof: at $\rho = 1$ all agents prove A0, 76% from their own facts (100% without a controller); at $\rho \leq 0.5$ at most 2%.

What the budget changes. A larger budget raises the false controller's end point by up to 0.13 ($\rho \leq 0.5$), up to 0.06 at $\rho = 0.75$ and nothing at $\rho = 1$, with each extra message worth about half as much. The gold standard's "the budget sets the end point and the rate sets the speed" becomes: *faster rates and larger budgets both raise the end point, with diminishing returns per message; the 30-message cap is what made the rate look irrelevant.*

3 Part B: a board that forgets faster or slower ($W = 24$, $W = 240$)

$W = 24$ (posts readable for about 1 time unit), the gold standard ($W = 48$, about 2) and $W = 240$ (about 10); the budget is $B = 30$ in all three.

3.1 Votes over time

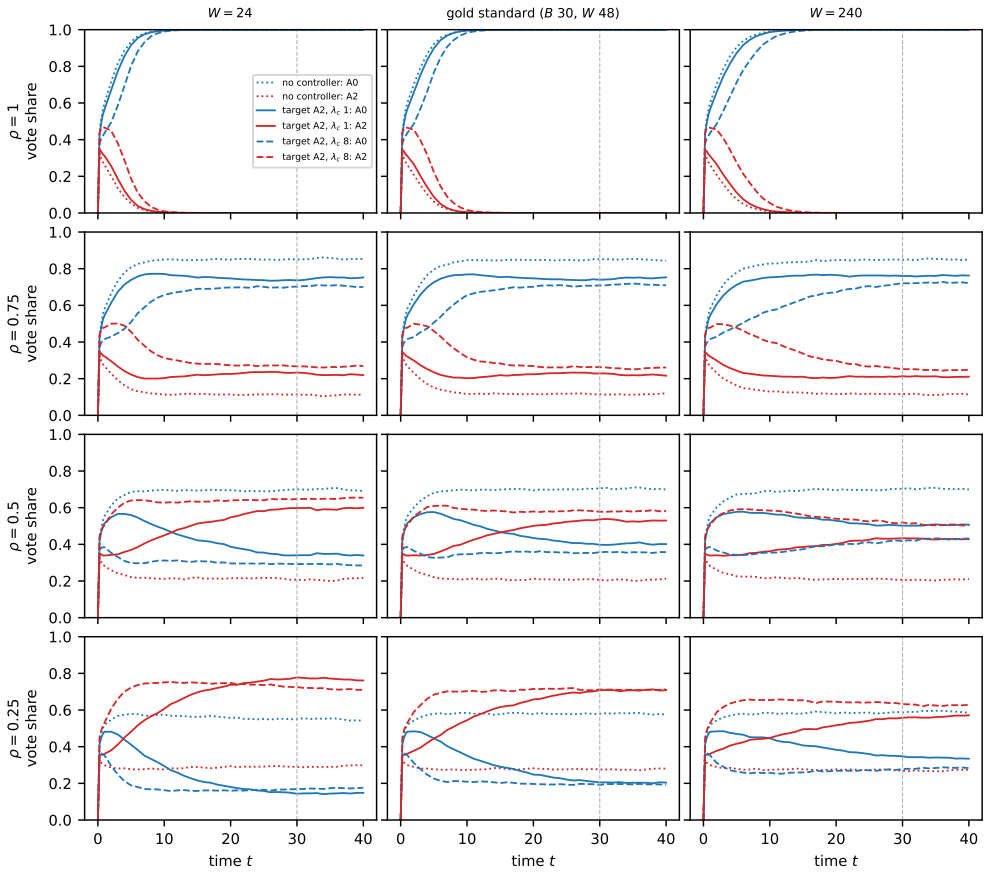


Figure 10: As Figure 1, for the three windows.

- **Without a controller the window hardly matters**: the shares on $t = 30$ differ by at most 0.03.
- **With full memory every window is immune**: A2 is back to 0 by $t = 30$.
- **A longer window protects**: A2 on $t = 30$ at $\lambda_c = 1$ is 0.60, 0.54 and 0.43 for $W = 24$, 48, 240 at $\rho = 0.5$, and 0.78, 0.71, 0.56 at $\rho = 0.25$; at $\lambda_c = 8$, 0.65, 0.58, 0.52 and 0.73, 0.71, 0.63.

3.2 Susceptibility over time, and per message

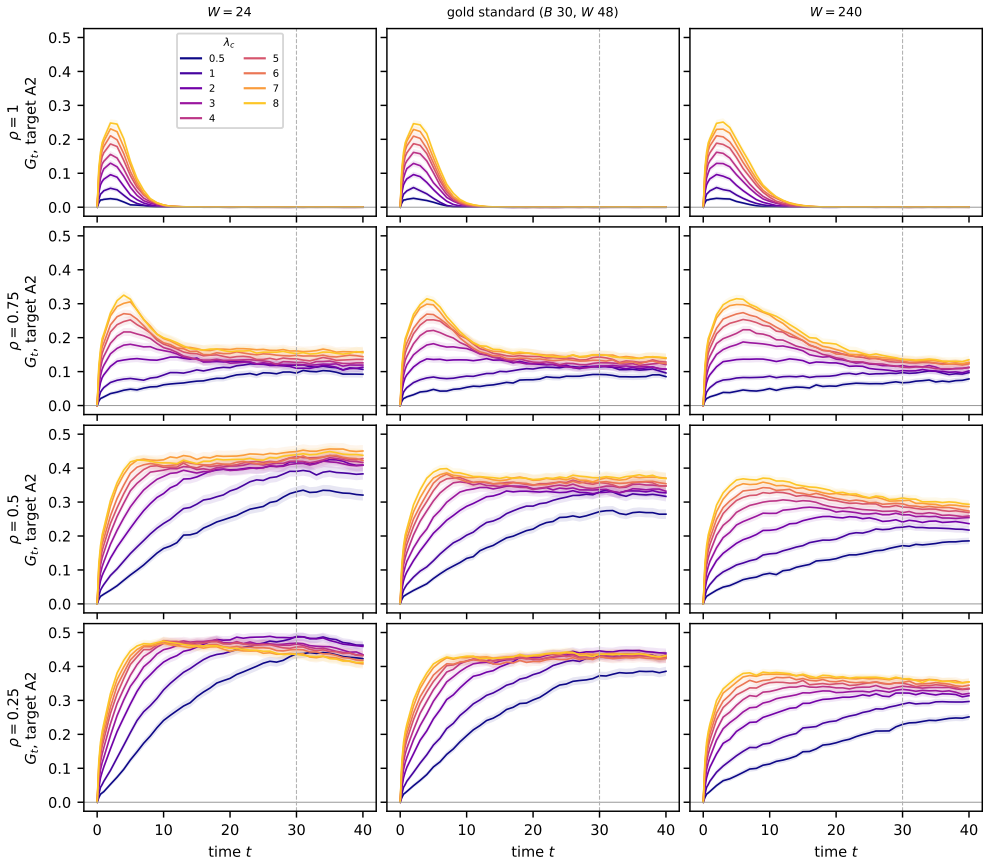


Figure 11: Paired susceptibility G_t, target A2, one curve per rate (bands: 95% intervals).

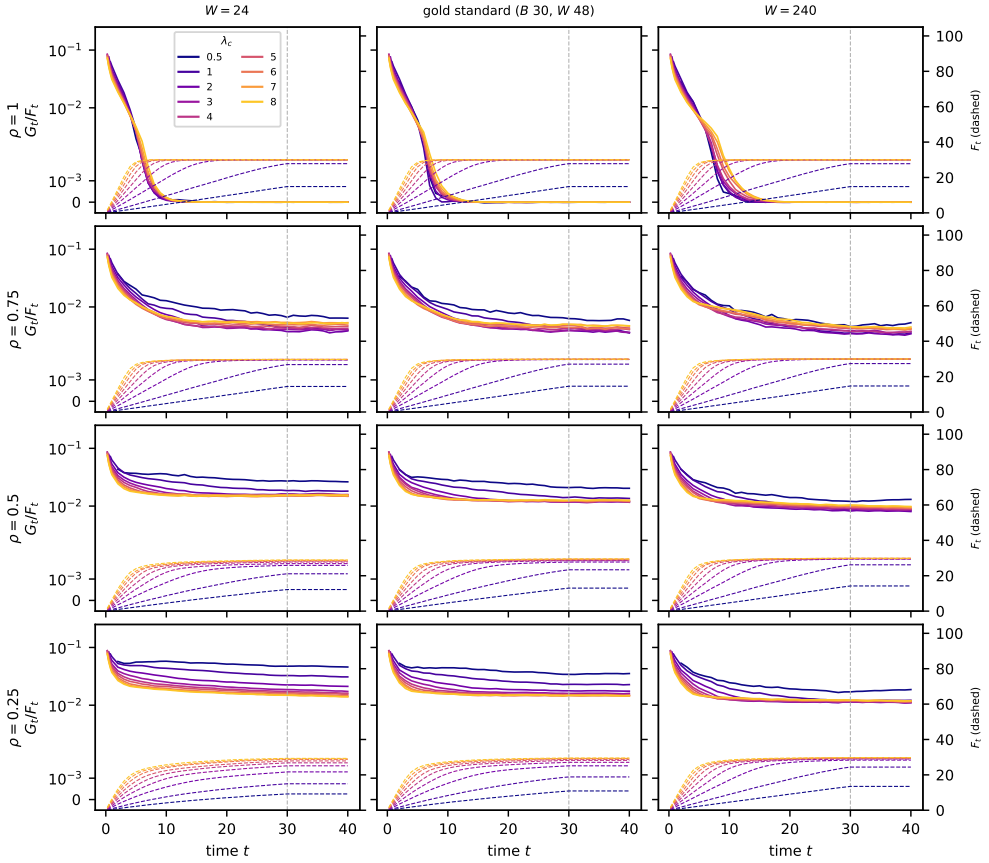


Figure 12: Gain per message G_t/F_t (solid) and messages posted F_t (dashed), target A2, one colour per rate.

- $W = 24$ **raises** G**,** $W = 240$ **lowers it, at every rate** (Figure 11): on $t = 30$, $\rho = 0.5$: 0.33–0.45, 0.27–0.37, 0.17–0.31 for $W = 24$, 48, 240; $\rho = 0.25$: 0.44–0.49, 0.37–0.45, 0.23–0.36. Rate by rate, $W = 240$ lowers G by 0.06–0.14 at $\rho \leq 0.5$ and $W = 24$ raises it by up to 0.09; at $\rho = 0.75$ both change it by at most 0.03. With $W = 240$ fast controllers peak early and fade ($\rho = 0.5$, $\lambda_c = 8$: 0.37 at $t = 5$, 0.31 at $t = 30$).
- **Readability follows** (Figure 14, middle): on $t = 30$, $\rho = 0.5$, $I = 0.20$–0.34, 0.15–0.27, 0.09–0.21 bits. **The efficiency does not move** (right): $\eta = 0.78$–0.93 at $\rho < 1$ for every window.
- **Each message is worth more when the board forgets fast** (Figure 12): at $\lambda_c = 1$, $\rho = 0.25$: 0.033, 0.023 and 0.012 per message.

3.3 Phase diagrams

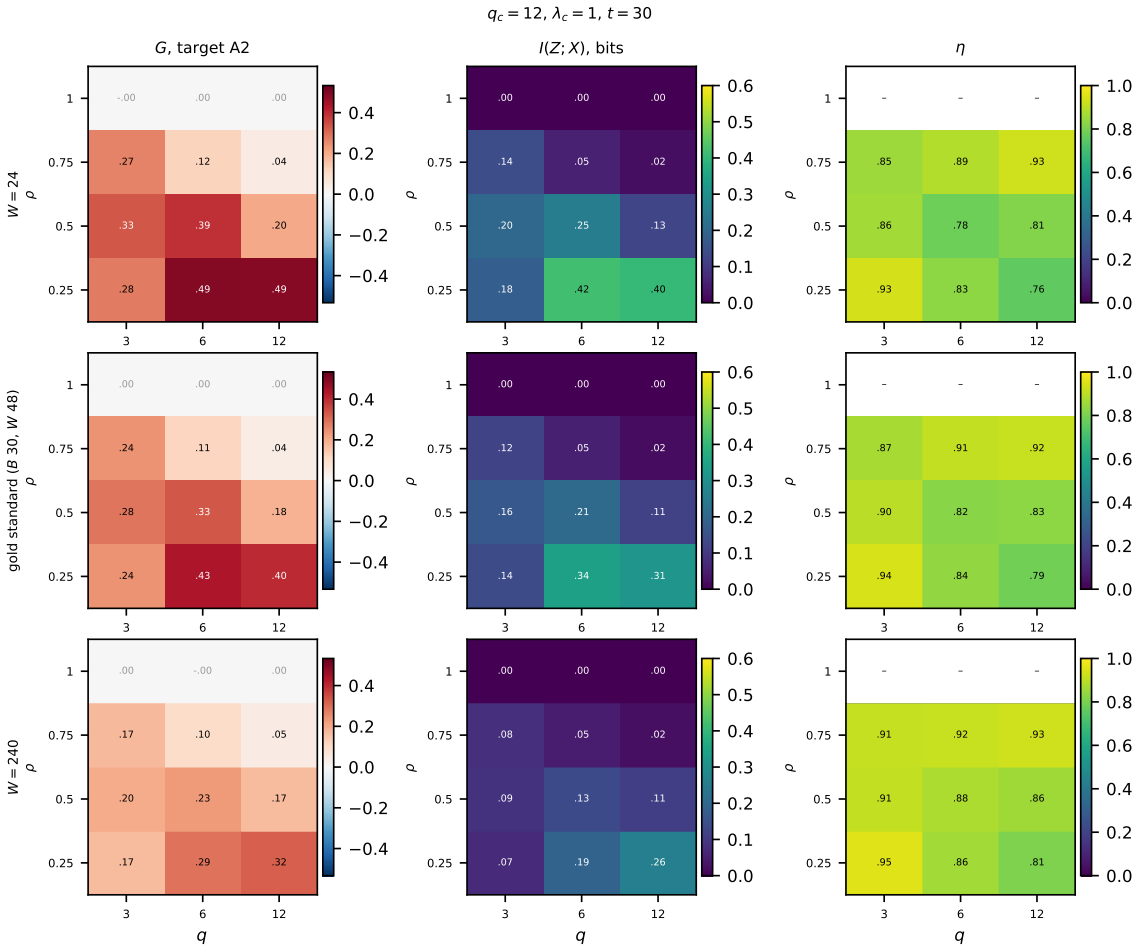


Figure 13: As Figure 4, for the three windows.

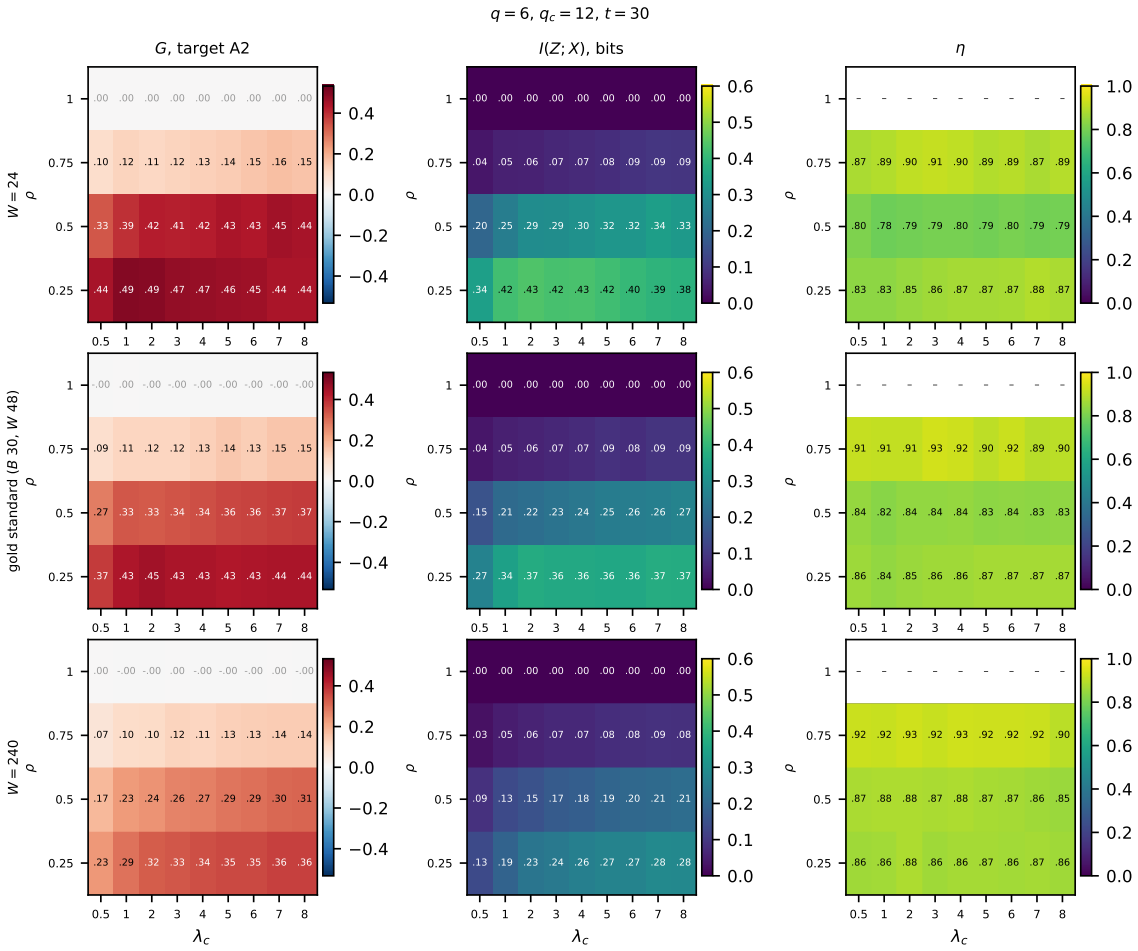


Figure 14: As Figure 5, for the three windows.

- **The reading reversal survives every window** (Figure 13), smaller with $W = 240$: from $q = 3$ to 12, G falls at $\rho = 0.75$ (0.27 → 0.04, 0.24 → 0.04, 0.17 → 0.05 for $W = 24$, 48, 240) and rises at $\rho = 0.25$ (0.28 → 0.49, 0.24 → 0.40, 0.17 → 0.32).
- **Memory stays the main axis** (Figure 14). $W = 240$ adds a weak rate gradient (from $\lambda_c = 1$ to 8: +0.09 at $\rho = 0.5$, +0.07 at $\rho = 0.25$); $W = 24$ none (+0.05, −0.05).

3.4 Susceptibility against the rate, against messages, and its slope

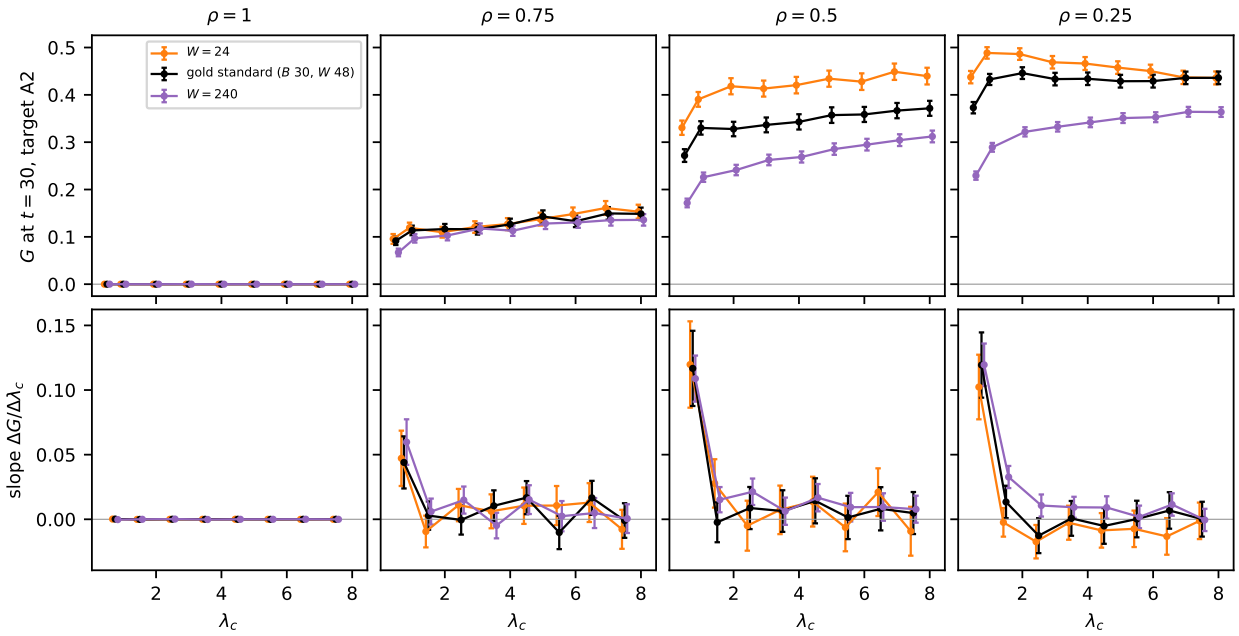


Figure 15: As Figure 6, for the three windows.

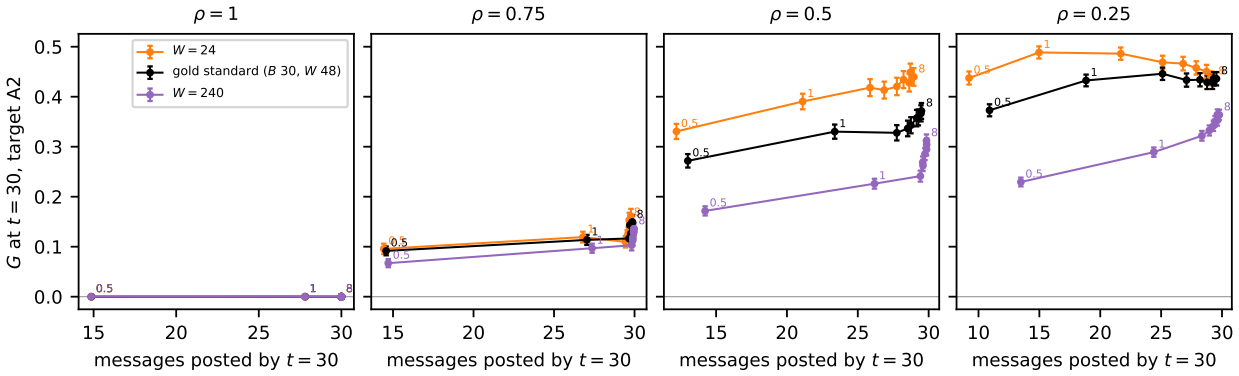


Figure 16: As Figure 7, for the three windows.

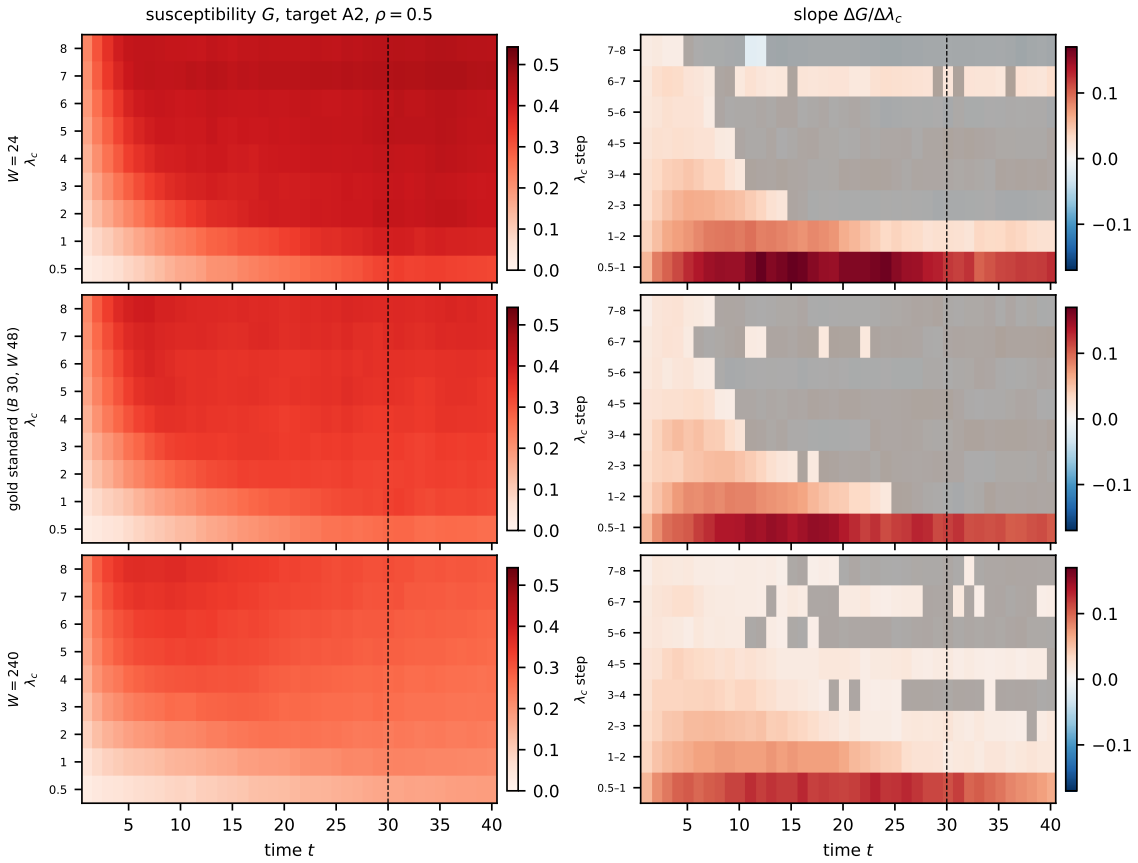


Figure 17: As Figure 8, for the three windows.

- **The window moves the whole curve** (Figure 16): every window posts about the same messages, but at the same 28–30 messages G is 0.45 ($W = 24$), 0.37 ($W = 48$) and 0.31 ($W = 240$) at $\rho = 0.5$.
- **The slopes look like the gold standard's** (Figures 15, 17): the first step dominates (+0.10 to +0.12 at $\rho \leq 0.5$); with $W = 240$ a few later steps are also slightly positive (up to +0.03).
- Gate off minus on: −0.03 to +0.03 for every window.

3.5 Who holds the proof

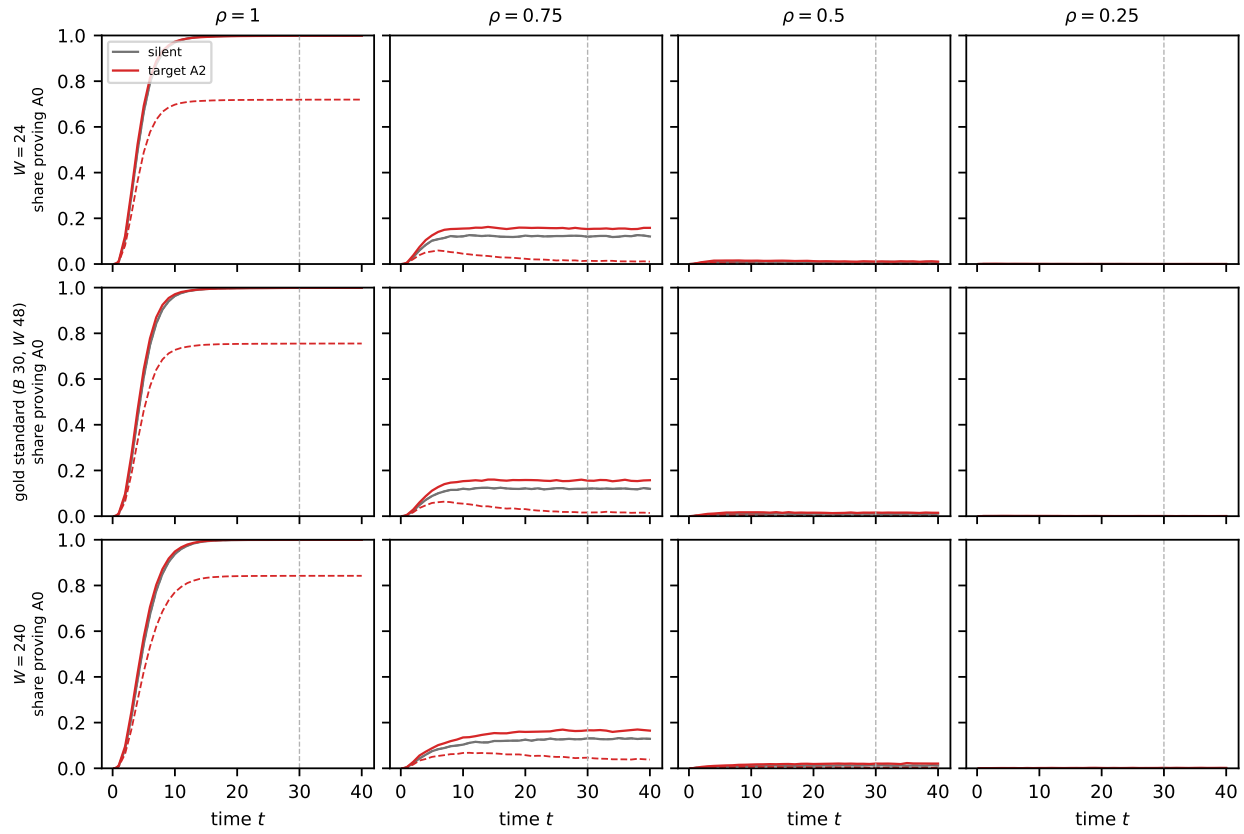


Figure 18: As Figure 9, for the three windows.

Only full memory keeps the proof, for every window. At $\rho = 1$ the window changes how many proofs stay the agents' own under the false controller: 72% ($W = 24$), 76% ($W = 48$), 84% ($W = 240$).

What the window changes. The board's memory protects like the agents' memory, but less: keeping posts readable for 10 time units instead of 2 lowers the false controller's gain by 0.06–0.14 at $\rho \leq 0.5$, keeping them 1 time unit raises it by up to 0.09, while the agents' memory ρ moves it by up to 0.43. A likely reason, not tested: a long window keeps the agents' older posts, which carry the facts of the proof, readable for longer.

4 Conclusions

1. **The main results of the gold standard hold in all three variants**: full memory makes the swarm immune; forgetting makes it steerable; reading more protects when agents remember and helps the controller when they forget; the gate matters little.
2. **The budget**: more messages raise the end point with diminishing returns, only where the swarm is already steerable (Part A).
3. **The board's memory**: a longer window protects the swarm, a shorter one exposes it; smaller effects than the agents' memory (Part B).
4. **The controller targeting A0** is left out of this report: in every variant its gain on $t = 30$ stays within 0.03 of the gold standard's, because the vote gate keeps it mostly silent (1–29 messages by $t = 30$).

Caveats. The window changes what the controller reads too: with $W = 240$ its vote gate sees votes up to 10 time units old, and at $\lambda_c = 1$ it posts 24–28 messages instead of 19–28. The variants were run at $q_c = 12$ and four memory values only, so the gold standard's $q_c \times \lambda_c$ diagram has no counterpart. One world, 24 exact-Bayesian agents, one controller policy; 1,000 episodes per setting (95% intervals on G about ± 0.02). Data: `results/simulation_2/2026-10-18_sim2_symmetric_*`; tables: `results/gold_standard/2026-10-18/` (variants) and `2026-10-16/` (gold standard); code: `shared_analysis/gold_standard/make_report_figures_2026-10-19.py`.